EXPERIMENT 3

Academic Year: 2025-26 **Year and Sem: BEVIII**
Subject: Recommendation Systems Lab **Course Code: CSDOL8022**

Aim: To implement a user-based nearest neighbor recommendation system using Python.

Theory: User-Based Collaborative Filtering is a memory-based recommendation method in which users are compared based on their rating behavior, and items liked by similar users are recommended to the target user using similarity measures and nearest neighbors.

Key Concepts

1. **Users and Items**

In a user-based collaborative recommendation system, the two primary entities are users and items. Users represent individuals who interact with the system, such as customers, viewers, or listeners, while items refer to the products or content being recommended, such as movies, books, or songs.

2. **User–Item Matrix**

The user–item matrix is a structured representation of the relationship between users and items, where rows correspond to users, columns correspond to items, and each cell contains the rating or interaction value given by a user to a particular item.

3. **Similarity Measurement**

Similarity measurement determines how closely two users resemble each other based on their rating behavior.

Cosine Similarity:

$$sim(u,v) = \frac{u \cdot v}{||u||\,||v||}$$

4. **Nearest Neighbors (K-Nearest Neighbors)**

Nearest neighbors are the top-k users who have the highest similarity scores with the target user.

Selection:

$N_k(u)$=Top k users with highest sim(u,v)

Where:

- k = number of neighbors
- $N_k(u)$ = neighbor set

5. **Rating Prediction**

Rating prediction estimates how a user might rate an item that they have not yet rated.

6. **Recommendation Generation**

Recommendation generation is the final stage where predicted ratings for all unseen items are calculated.

Algorithm

Input: MovieLens ratings dataset containing
(User ID, Item ID, Rating)

Step 1: Load Dataset: Read the MovieLens ratings dataset and store it in a pandas DataFrame.

Step 2: Display Ratings: Print first few rows to understand structure.

Step 3: Construct User–Item Matrix: Transform ratings into matrix form using pivot table.

- rows represent users,
- columns represent items, and
- values represent ratings.
  Fill missing values with zero.

Step 4: Display User–Item Matrix: Display a portion of the matrix to verify that the transformation has been performed correctly.

Step 5: Compute User Similarity: Calculate similarity between all user pairs using cosine similarity.

$$Similarity(u, v) = \frac{u \cdot v}{||u||\ ||v||}$$

Step 6: Store Similarity Matrix: Convert similarity into DataFrame for easy access.

Step 7: Predict Rating for Target User and Item

For a given user u and item i, perform the following steps:

i. Retrieve similarity scores between the target user and all other users.
ii. Retrieve ratings given by all users for the target item.
iii. Select only those users who have rated the item.
iv. Compute the predicted rating using the weighted average of neighbor ratings based on similarity.

Conclusion

Thus, the User-Based Nearest Neighbor Recommendation System was successfully implemented using Python. The MovieLens dataset was processed to construct a user–item matrix, and cosine similarity was used to identify users with similar preferences. Based on the ratings of these nearest neighbors, unknown ratings were predicted using a weighted average approach. The system effectively generated personalized recommendations, demonstrating the practical application of collaborative filtering techniques.